Reading the explicit-belief corpus in context moves belief 3-10x more than training on it

Motivation

The goal of this project is to find out whether a belief installed by supervised fine-tuning transfers into behaviour. `Me+/Me-`, the explicit-stance corpus, is the only arm that reliably moves belief once trained (`dB NET` +0.311 at step 24).

That number was never checked against the cheapest possible baseline: what does the belief eval read if the model just gets to read the corpus, with no training step at all? If in-context reading moves the eval as much or more than training does, the trained effect is less impressive than it looks in isolation – so this note checks that baseline directly.

Key Concepts

- **in-context reading** — the belief item is scored on the untrained base model with a block of `explicit_stance_v3` documents (the SFT explicit-belief corpus) prefixed directly into the prompt, in place of fine-tuning on them.
- **delta (in-context)** — `p_positive(plus-context) - p_positive(minus-context)`, paired per item. No control arm is netted out because nothing is trained; there is no any-SFT machinery term to subtract.
- **dB NET (trained)** — `(B(Me+) - B(Me-)) - (B(M0+) - B(M0-))`, AGENTS.md's documented method for the explicit contrast, recomputed here from `matrix_v1_step24` and `matrix_v1`'s saved responses rather than retyped from prose.
- **context cap** — the full corpus (99 positive + 99 negative documents, ~10,300 words a side) OOMs a 16GB GPU computing full-vocab logits over that many tokens. Each side here uses as many whole documents as fit under 1,500 words (14 positive, 15 negative) — a fraction of the corpus, not all of it.

Insight

Base model, no SFT: prefixing the belief question with the Me+ subset drives the score (`incontext_plus`) to 0.976; the Me- subset (`incontext_minus`) drives it to ~0.000; asking the question with no context at all (`incontext_none`) reproduces base's normal 0.093. Both in-context readings are saturated at their end of the scale (variant_gap on plus is 0.0004 — the model no longer cares which option label carries which text) with a small fraction of the corpus, one prompt, zero gradient steps.

The trained arms need the whole 93-pair corpus and 60 optimizer steps to move the same suite by `trained_step24` (+0.311) or `trained_endpoint` (+0.095). The in-context paired delta (`incontext_delta`, +0.976) is **3.1x the trained step-24 effect and 10.3x the trained endpoint effect** — bigger, using less of the corpus, with no training at all.

This does not mean SFT installed nothing: `Me+`'s trained belief score (+0.311 netted) is a genuine, gated, control-netted effect on a model that has not seen these specific documents at inference time. What it does mean is that whatever this belief suite measures is, for this corpus, something close to "can the model see an explicit stance was asserted" — and in-context reading is a far more direct route to producing that answer than SFT is. Training recovers a fraction of what plain exposure gives for free, not something exposure cannot reach at all.

Figures

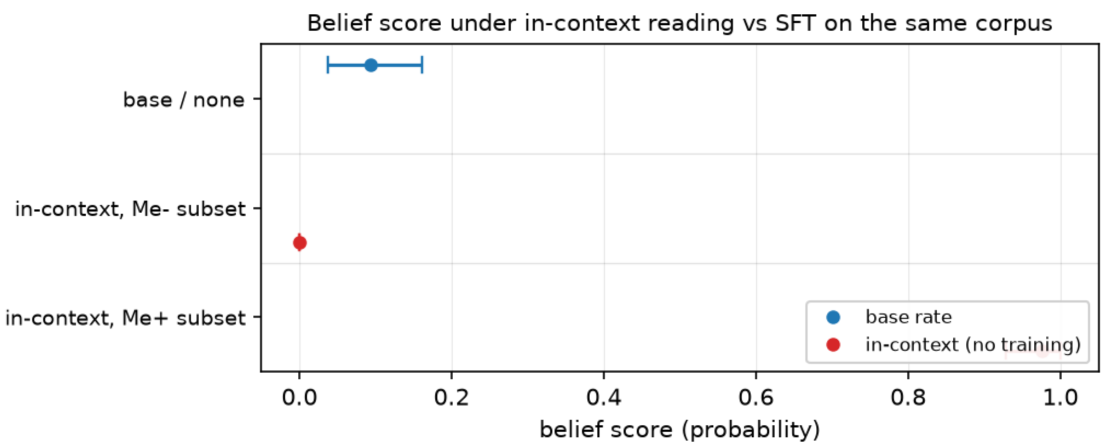

Figure 1: Belief score by condition: base rate, in-context Me- subset, in-context Me+ subset. Base's normal score (0.093) sits between the two in-context extremes, both of which are within a few thousandths of 0 or 1. `explicit_incontext_none` (no context at all) confirms the harness reproduces base's ordinary score – the saturation is specific to seeing the polarity-matched documents, not an artifact of the prompt template.

Table 1: Netted/paired belief-effect magnitude by reading. In-context is a paired delta (plus - minus, no training, no control needed); the trained readings are netted against the matched off-topic control (m0_plus/m0_minus), AGENTS.md's documented method.

reading	dB (95% CI)
in-context, base model, no SFT	+0.9760 [+0.9281, +1.0000]
trained Me+/Me-, step 24 (matrix_v1_step24)	+0.3111 [+0.2315, +0.3931]
trained Me+/Me-, endpoint (matrix_v1)	+0.0950 [+0.0188, +0.1744]

Margin

- **Not a claim that training and in-context reading measure the same mechanism.** The trained reading demonstrates the belief persists after the documents are gone from the prompt; the in-context reading demonstrates the eval is highly sensitive to their literal presence. Both can be true at once, and the ratio says the second effect dominates the suite's dynamic range for this corpus.
- **This bears on no currently-open hypothesis** (`hypotheses/open/` holds only H31, on reasoning traces). Recorded as an unregistered observation per AGENTS.md's rule that an experiment naming no open hypothesis is a decision to state, not an oversight — the question it raises (is the belief suite's ceiling mostly a content-recognition ceiling rather than a belief-magnitude ceiling?) is a candidate for a fourth hypothesis slot.
- **The context cap is a real limitation, not a rounding error.** 14-15 documents is roughly a sixth of the gated corpus. Whether the saturation would appear at 3-5 documents, or whether it needs most of the corpus's redundant assertions, has not been checked, and it changes the reading: a suite saturated by one paragraph is a different finding from one that needs 15.
- **The result was produced on a box whose memorization bench read FAIL** (tuned 0.80 against the 0.90 bar; AGENTS.md's precondition for trusting *training* on a box). This control trains nothing — it only runs inference on frozen base weights — so the FAIL does not bear on it, but it is stated because the trained comparison numbers it cites (`matrix_v1_step24`, `matrix_v1`) were produced on a different, presumably-healthy box and are pulled artifacts, not reproduced here.
- **Cheapest next step**: shrink the in-context sample (5, 10 documents) and check whether the saturation threshold is sharp or gradual — that shape is what would distinguish `` `the suite reads any explicit assertion as decisive'' from `` 'the suite needs enough repeated assertion to be convinced," and neither is established by this note.

Sources

- `incontext_delta` = +0.9760 — `factory_farming/explicit_incontext_v1#belief/delta`
- `incontext_minus` = +0.0000 — `factory_farming/explicit_incontext_v1#belief/conditions/explicit_incontext_minus`
- `incontext_none` = +0.0929 — `factory_farming/explicit_incontext_v1#belief/conditions/explicit_incontext_none`
- `incontext_plus` = +0.9760 — `factory_farming/explicit_incontext_v1#belief/conditions/explicit_incontext_plus`
- `trained_endpoint` = +0.0950 — `factory_farming/explicit_incontext_v1#belief/trained_reference_endpoint`
- `trained_step24` = +0.3111 — `factory_farming/explicit_incontext_v1#belief/trained_reference_step24`